

CS & IT ENGINEERING



Digital Logic

Combinational Circuits

Lecture No. 8



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Recap of Previous Lecture



Topic



MUX ✓

De-mux ✓

Encoder ✓

Priority Encoder ✓

Topics to be Covered



Topic

Combinational Circuits

Encoding (Intro)
Decoder (*)



Intro to Encoding:

- 1> Decimal
- 2> BCD $\rightarrow$ Binary Coded Decimal
- 3> Excess 3 Code
- 4> 4221
- 5> Grey Code

1) Decimal $\swarrow$ weighted code
 $\searrow$ Self-complemented codes

$\rightarrow [0 \text{ to } 9]$

$(10^3 \quad 10^2 \quad 10^1 \quad 10^0)$
 $d_3 \quad d_2 \quad d_1 \quad d_0$

$(352)_{10}$

$\rightarrow 3 \times 10^2 + 5 \times 10^1 + 2 \times 10^0$

Complement

$r \rightarrow$ Base (Radix)

$(r-1)$'s Complement

r 's Complement

$r=2$

1's

2's

$r=8$

7's

8's

$r=10$

9's

10's

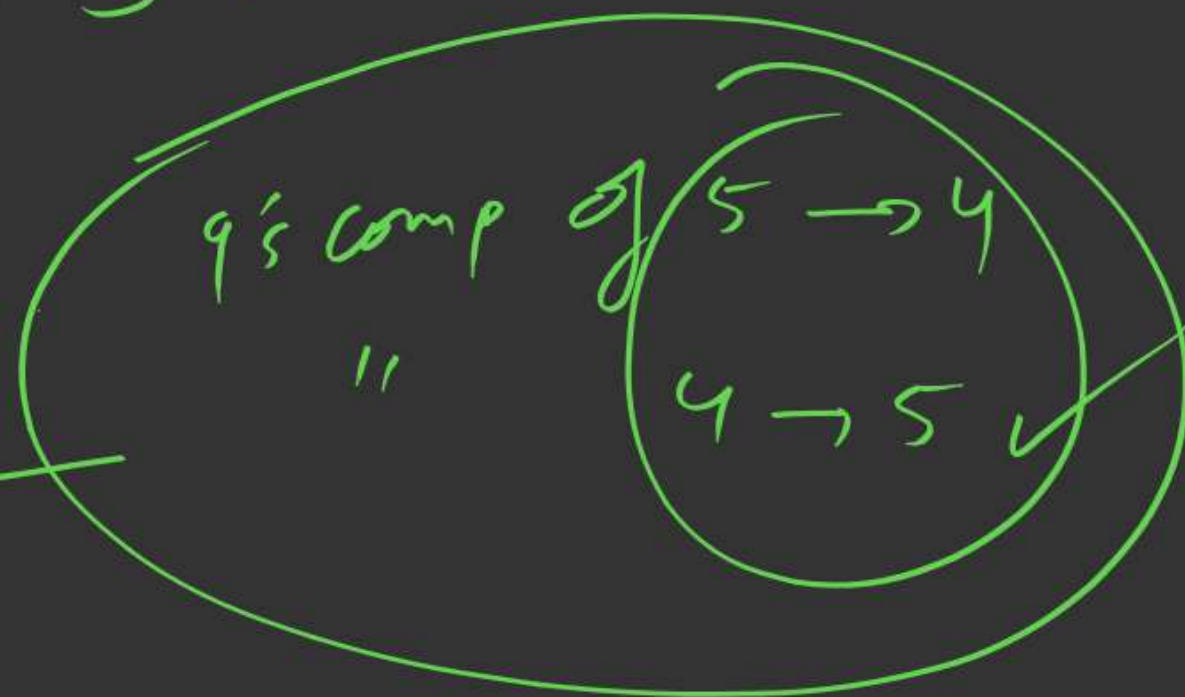
Decimal

* 9's Complement

1) 9's Complement of 4? $\Rightarrow 5 \checkmark$

$$\begin{array}{r} 9 \\ - 4 \\ \hline 5 \end{array}$$

self
Complemented
code



2) 9's " of 5? $\begin{array}{r} 9 \\ - 5 \\ \hline 4 \end{array} \Rightarrow 4 \checkmark$

2) BCD (Binary Coded Decimals)

↳ Each decimal number are represented
by 4 bits.

eg:- Decimal $\rightarrow 1$

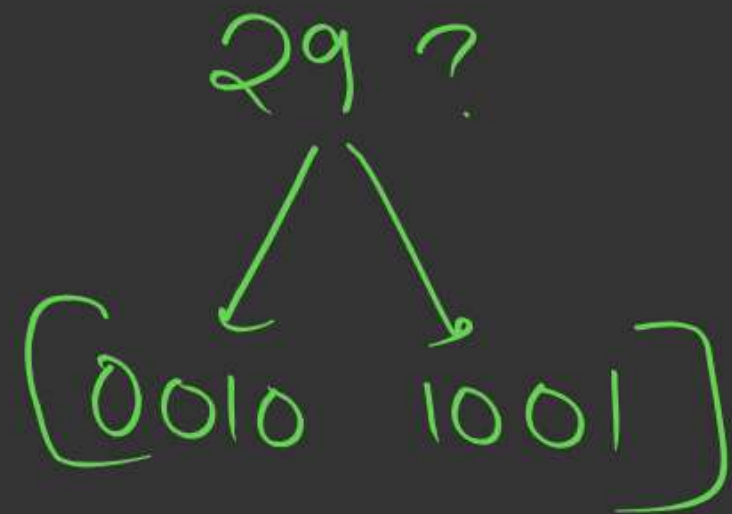
BCD $\rightarrow$ 0001

4
 $\downarrow$
0100

9
↓
1001

10
↓
1010 ? ~~X~~ BCD
↙
Binary ✓

10
↙ ↘
[0001 0000]
↓
00010000
BCD

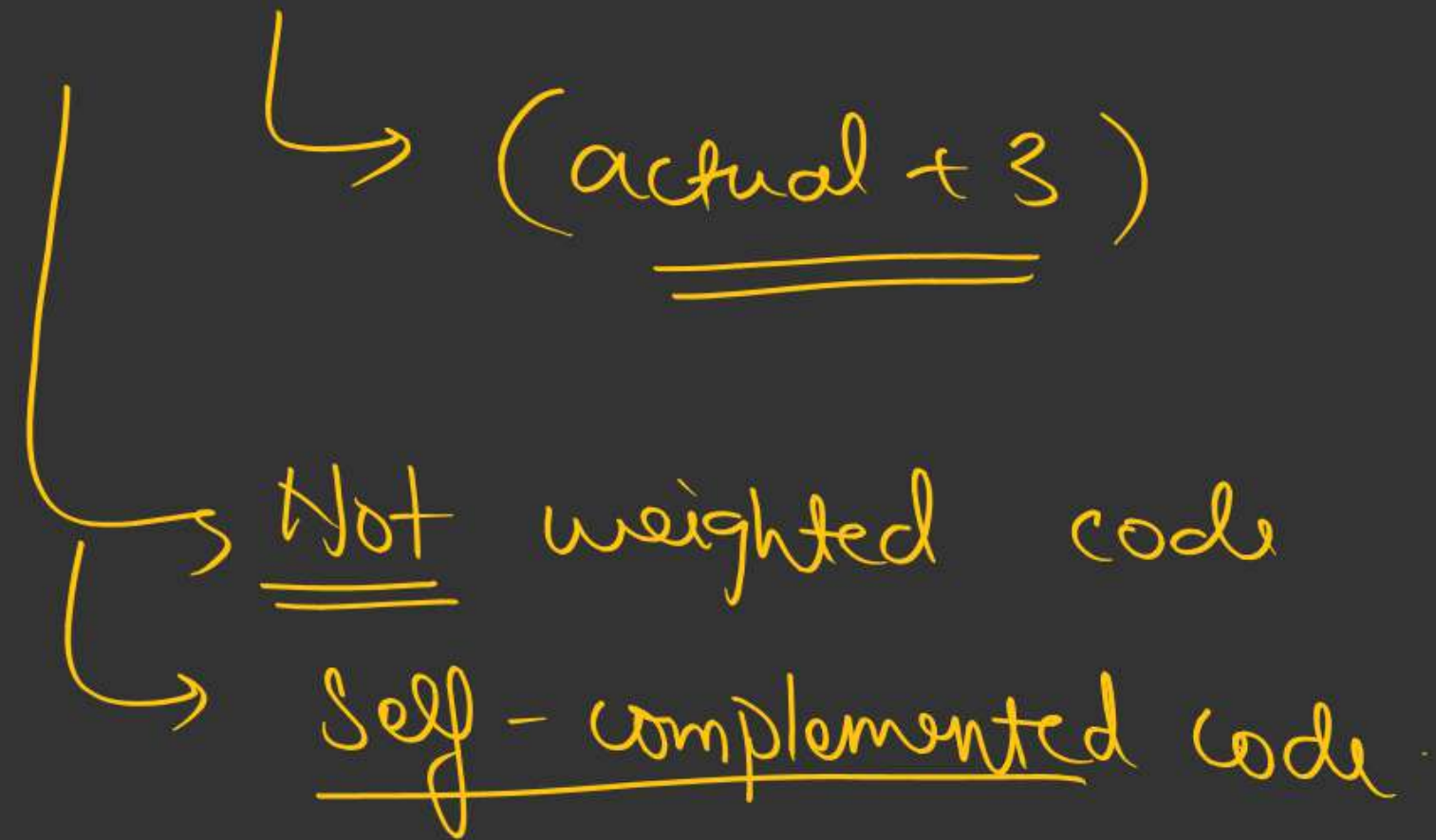


BCD

→ weighted code

→ (Not) self-complemented codes

3) Excess-3 code



4) 4 2 2 1 code

$\downarrow$
 $[4 \ 2 \ 2 \ 1]$
 $a_3 \ a_2 \ a_1 \ a_0$

$\rightarrow$ $\left[\begin{array}{l} \text{weighted} \\ \text{self-complemented} \end{array} \right]$

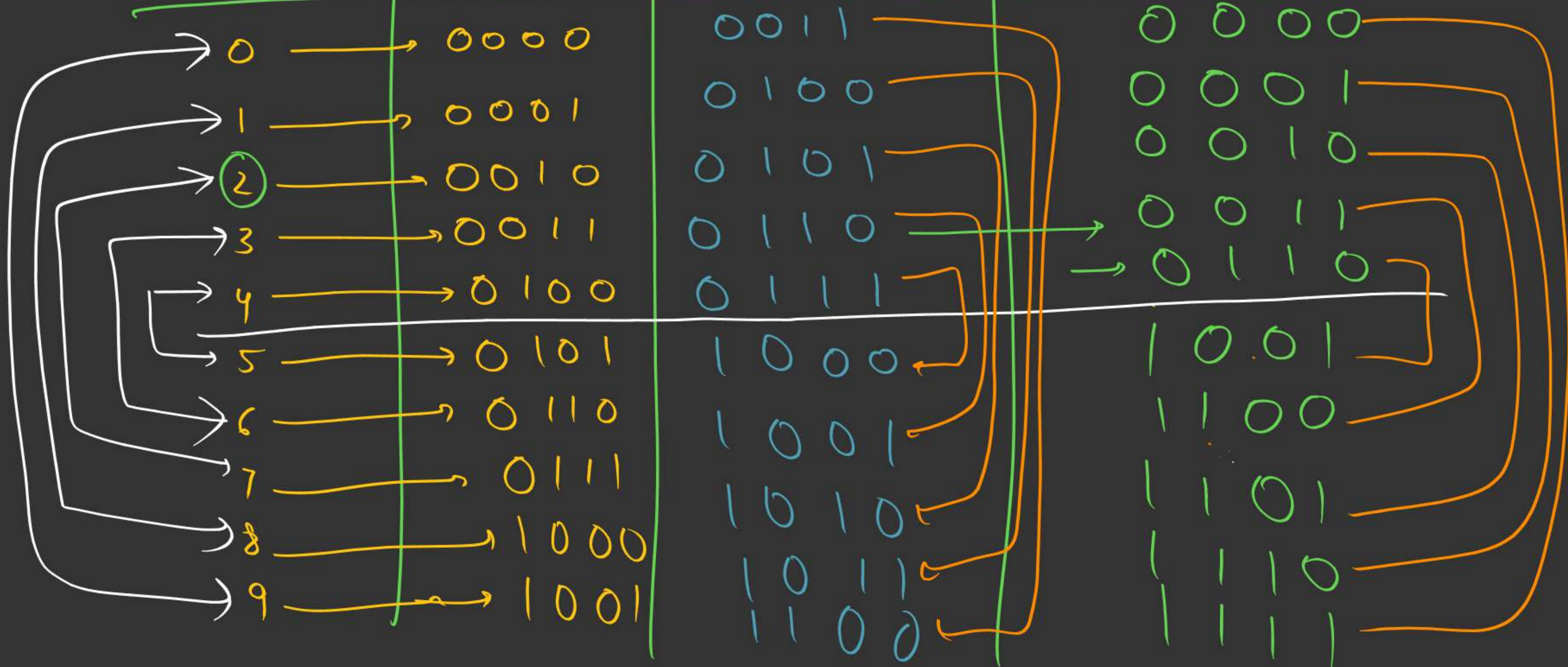
(+3) ✓

Excess-3

✓
4 2 2 1

✓
Decimal

BCD



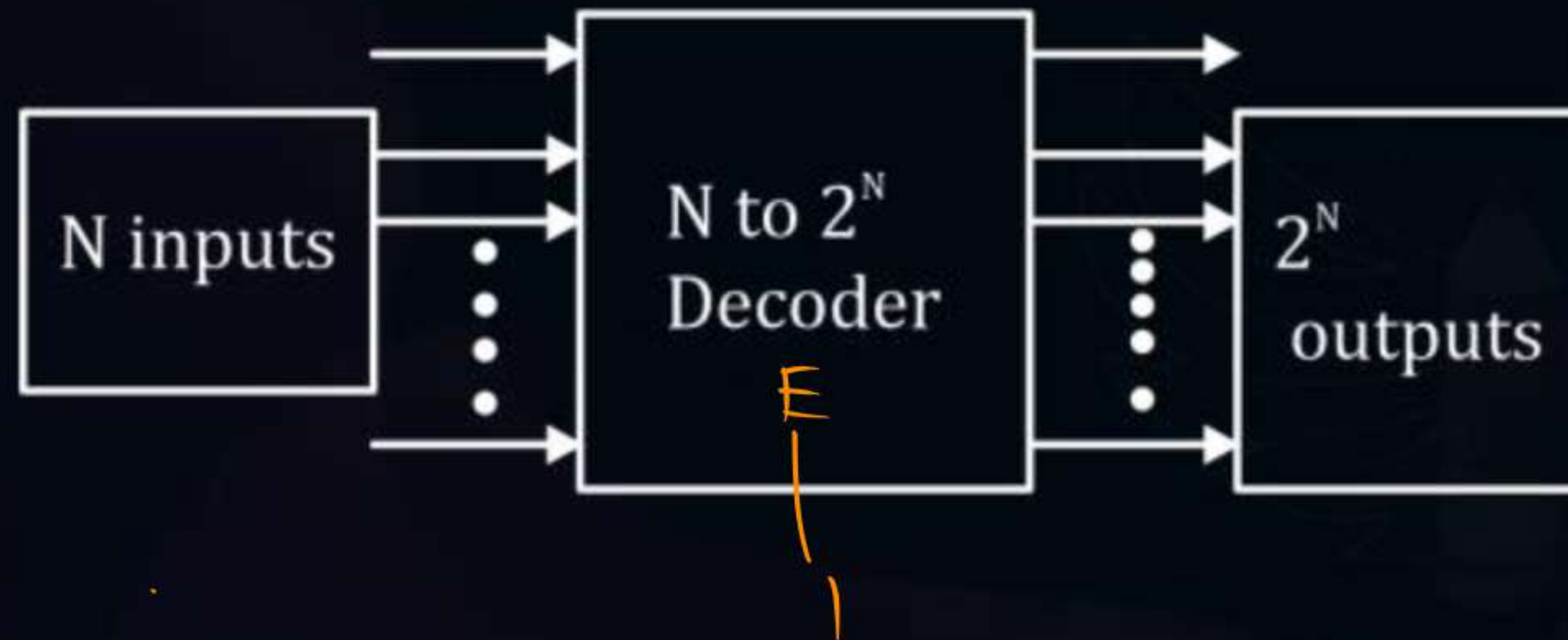


Topic : DECODER, HA FA

DECODER: Circuit which is use to convert binary into any other code.

1. 2 × 4 Decoder
2. 3 × 8 Decoder
3. 4 × 16 Decoder

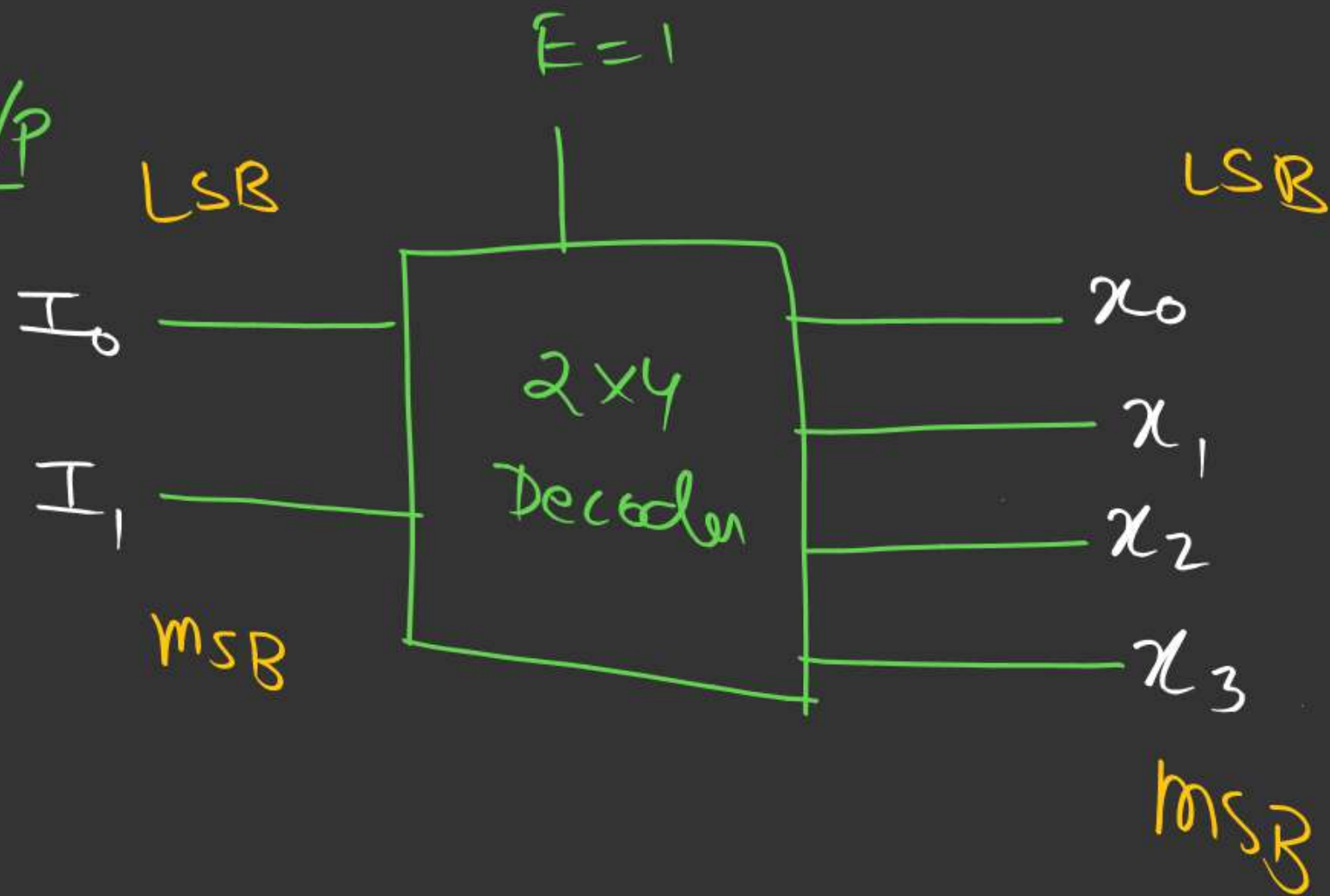
$2^n \times n$ Encoder



eg1:- Design 2x4 Decoder

Step 1:- i/p & o/p

Standard



Step 2:- Truth Table:

Input		O/P			
I_1	I_0	x_3	x_2	x_1	x_0
0	0	0	0	0	1
0	1	0	0	1	0
1	0	0	1	0	0
1	1	1	0	0	0

Step 3:- logical expression:

$$x_0 = \overline{I_1} \cdot \overline{I_0}$$

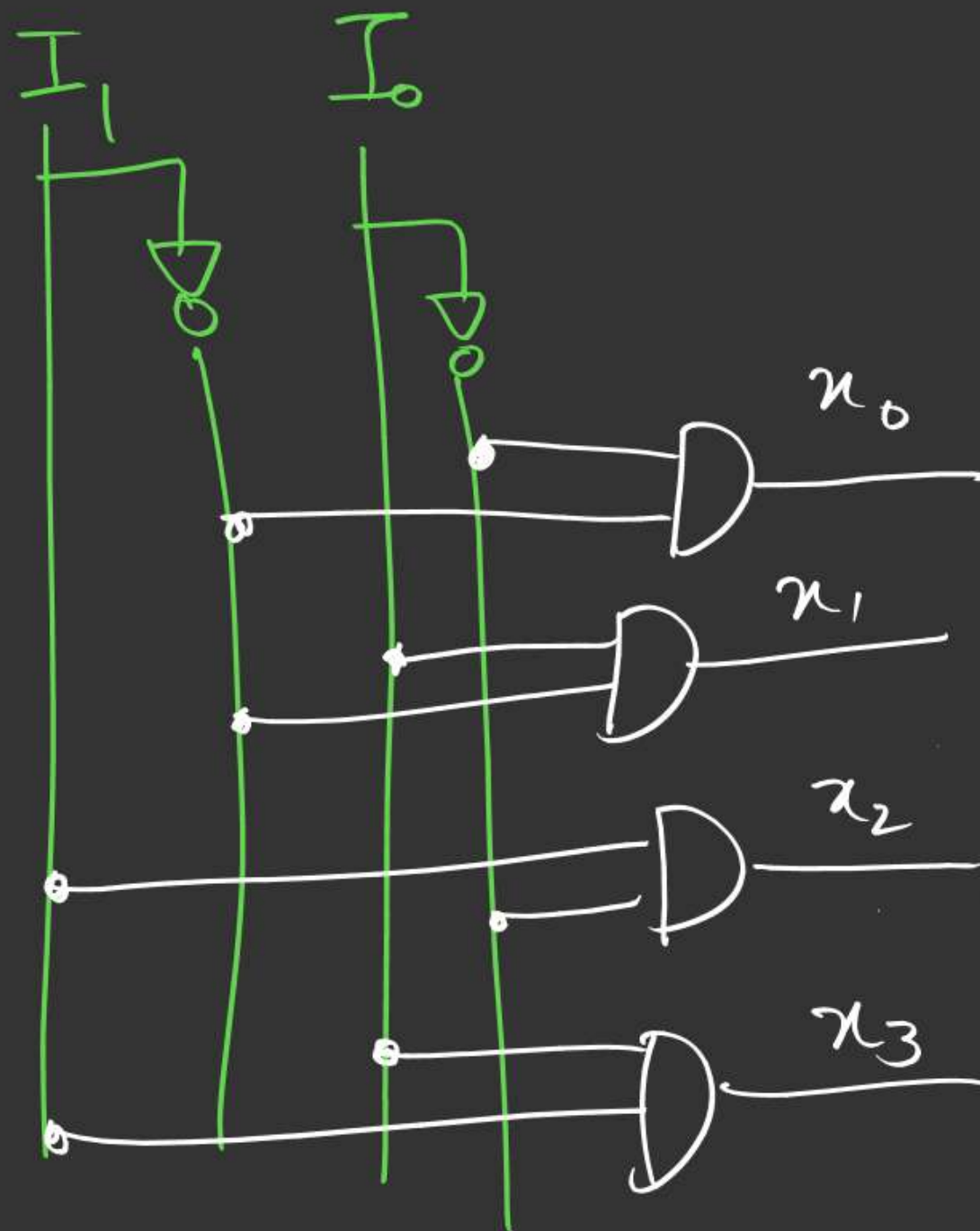
$$x_1 = \overline{I_1} \cdot I_0$$

$$x_2 = I_1 \cdot \overline{I_0}$$

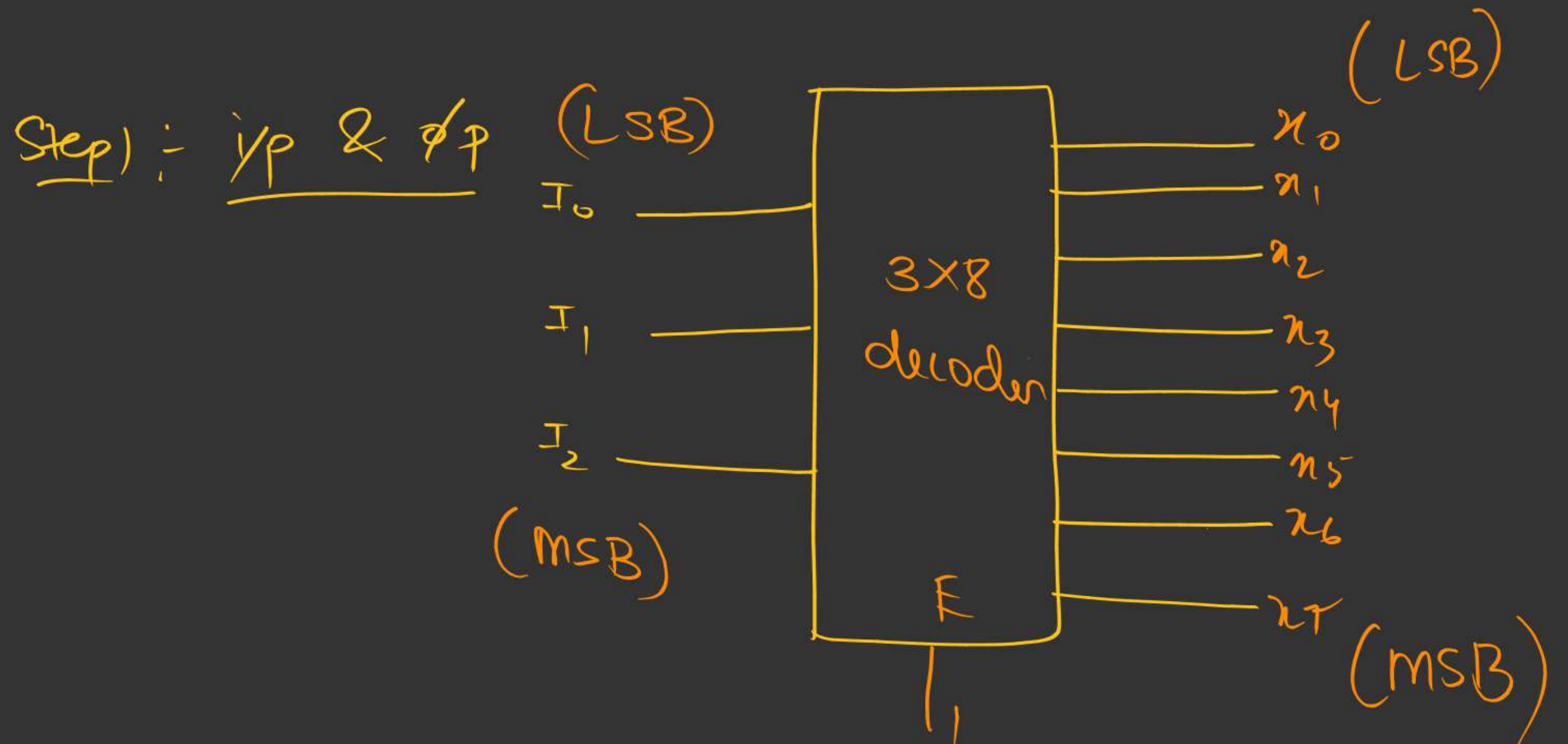
$$x_3 = I_1 \cdot I_0$$

Step 4:- Minimization
(not needed)

Steps :- Hardware design.



(Q.2) Design 3x8 Decoder:



Step 2: Truth Table:

Step 3: Logical Expression

$$x_0 = \bar{I}_2 \bar{I}_1 \bar{I}_0$$

$$x_1 = \bar{I}_2 \bar{I}_1 I_0$$

$$x_2 = \bar{I}_2 I_1 \bar{I}_0$$

$$x_3 = \bar{I}_2 I_1 I_0$$

$$x_4 = I_2 \bar{I}_1 \bar{I}_0$$

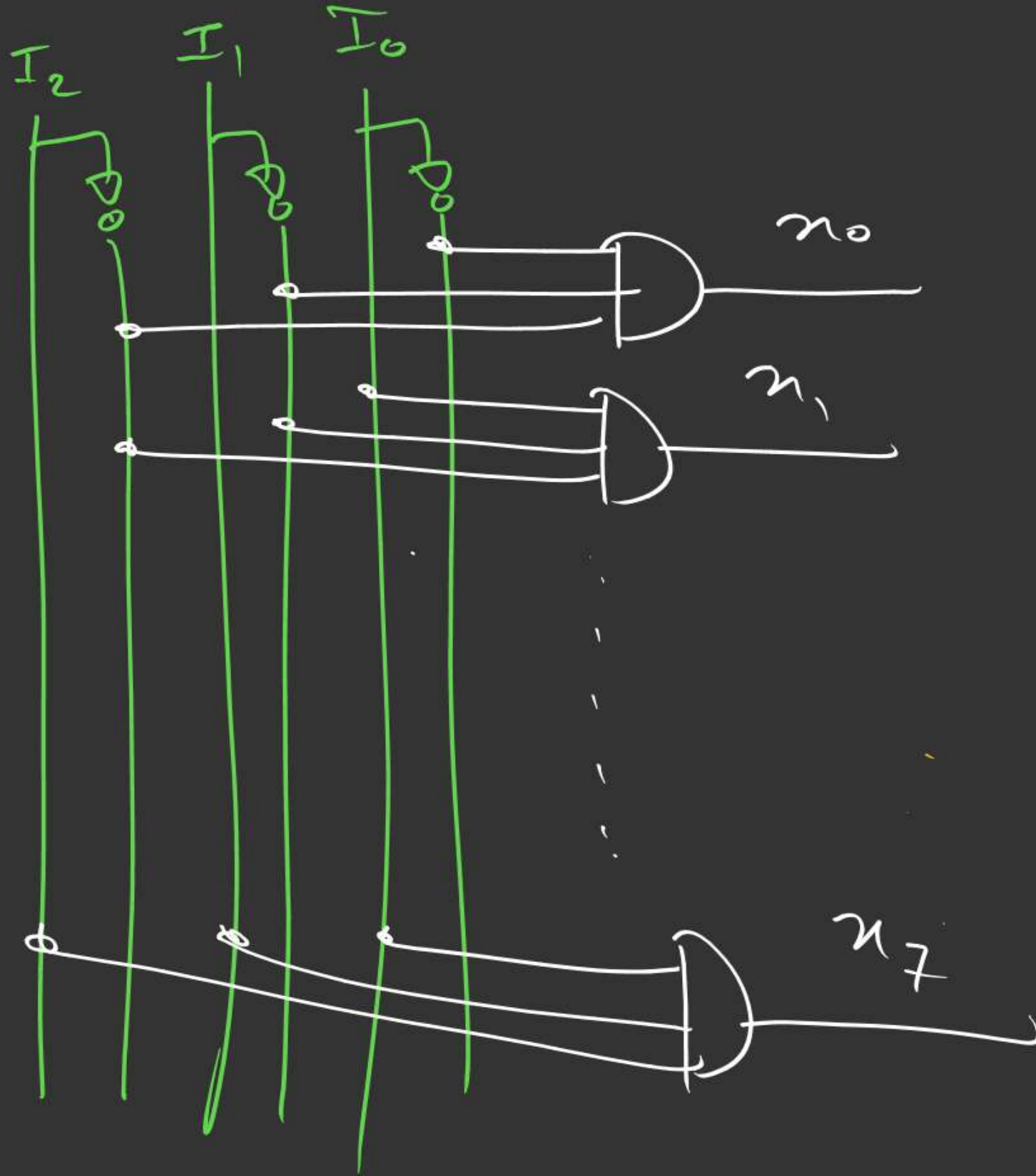
$$x_5 = I_2 \bar{I}_1 I_0$$

$$x_6 = I_2 I_1 \bar{I}_0$$

$$x_7 = I_2 I_1 I_0$$

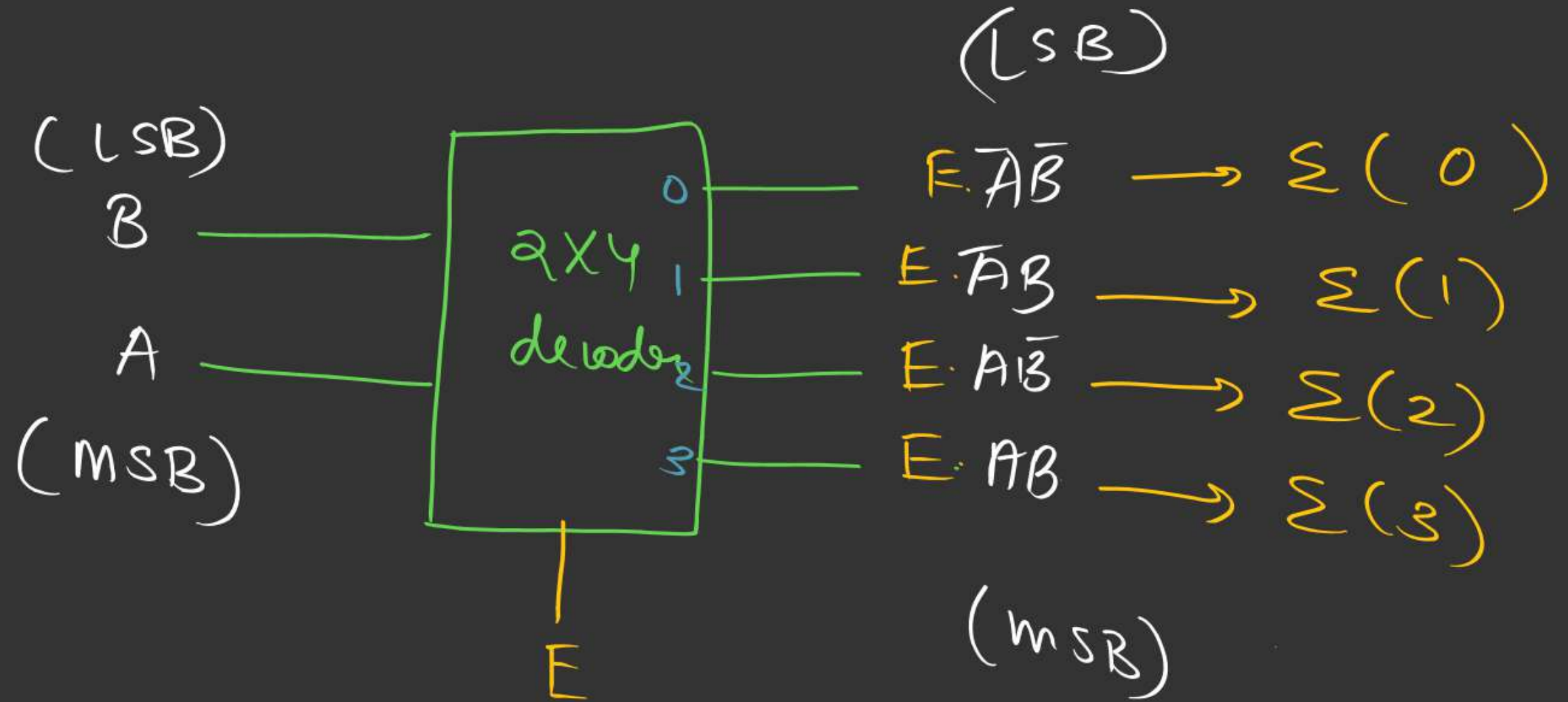
Step 4: Minimization

↳ already done



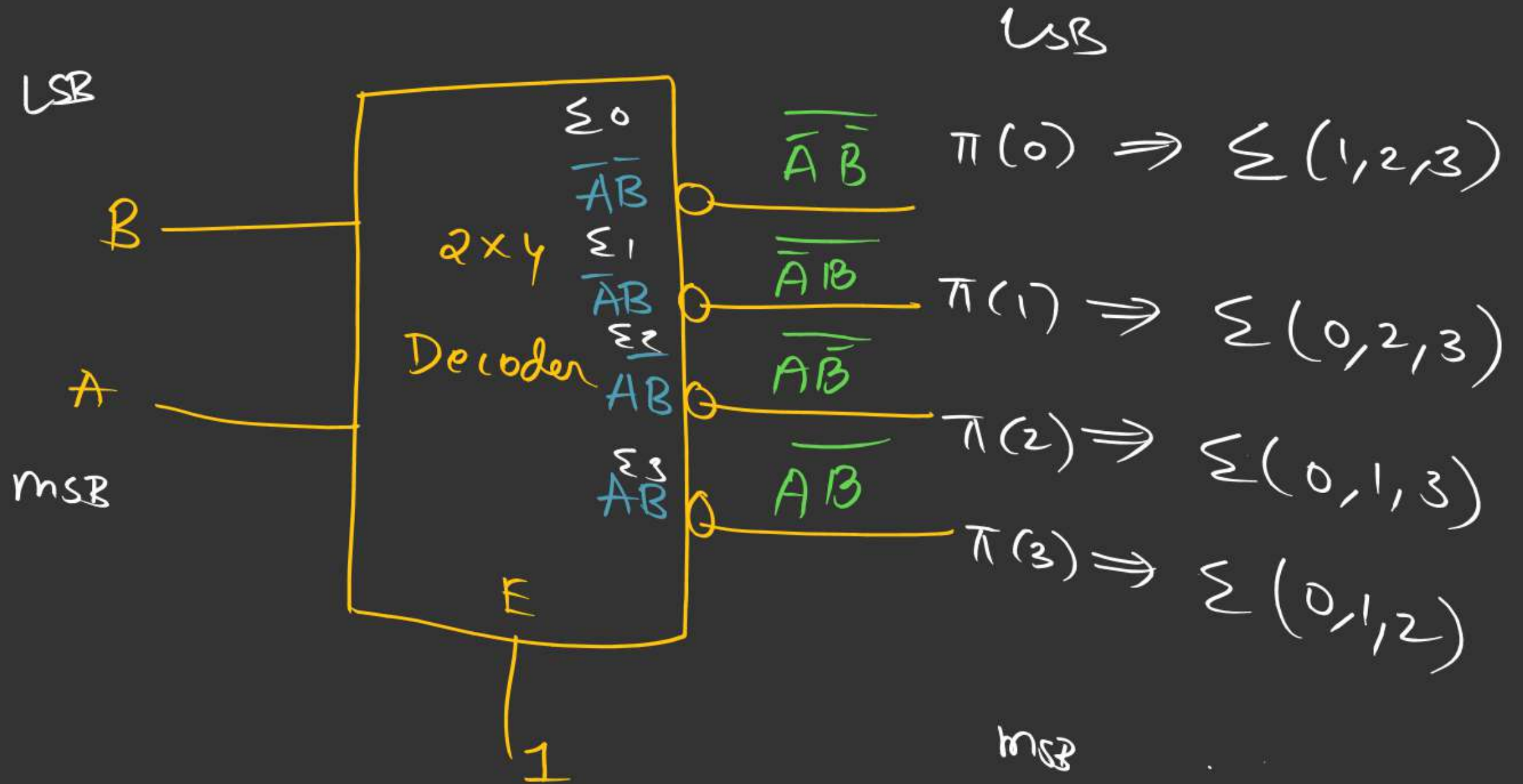
HW : Design 4X16 Decoder.

Concept (Standard)



(eq 2)

$$f(A, B) = \sum (0) \\ = \pi (1, 2, 3)$$



$$\overline{A \cdot B} = \overset{0 \ 0}{A + B} \Rightarrow \pi(0)$$

$\hookrightarrow \Sigma(1, 2, 3)$

A	B	A + B
0	0	0
0	1	1
1	0	1
1	1	1

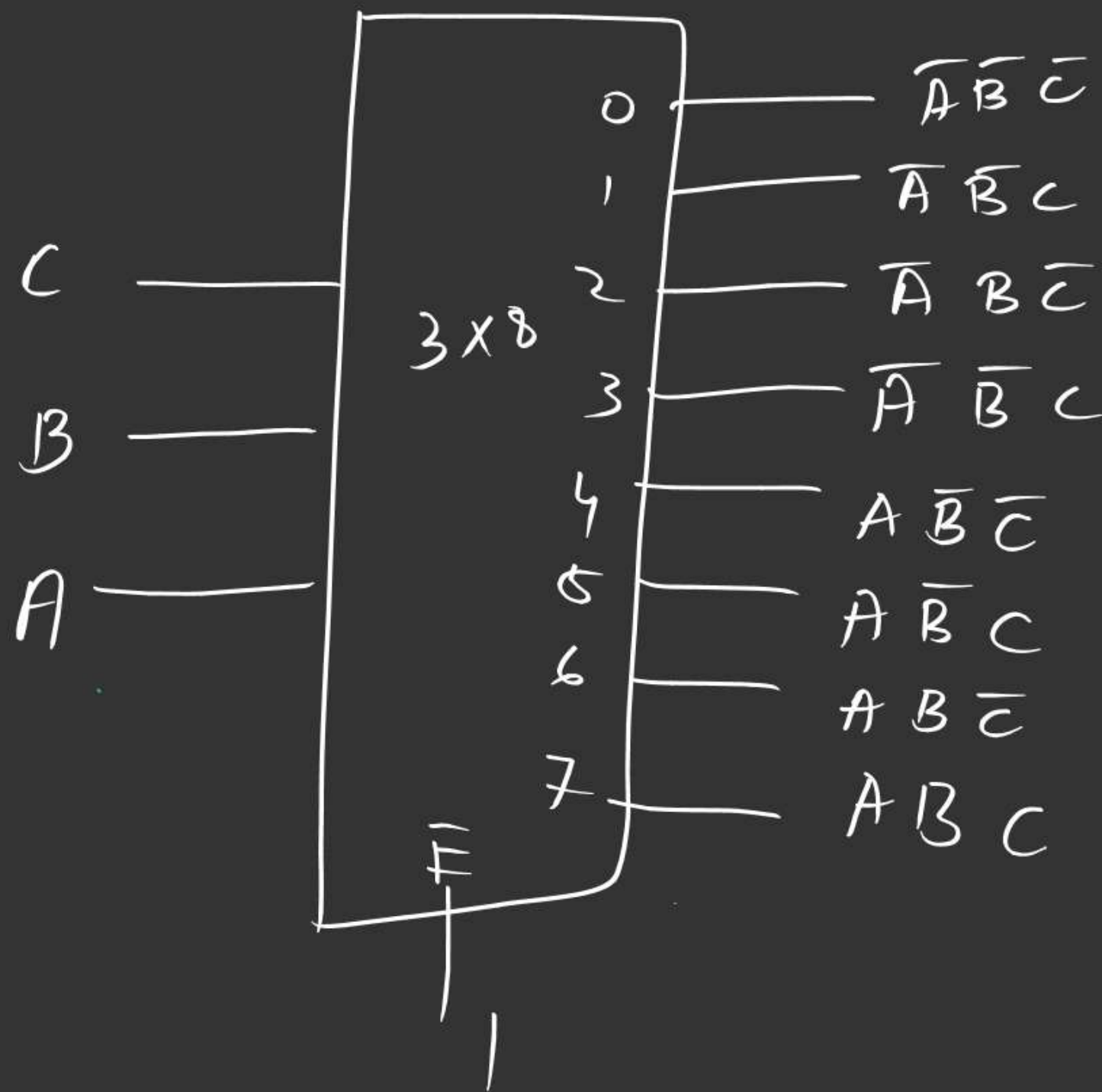
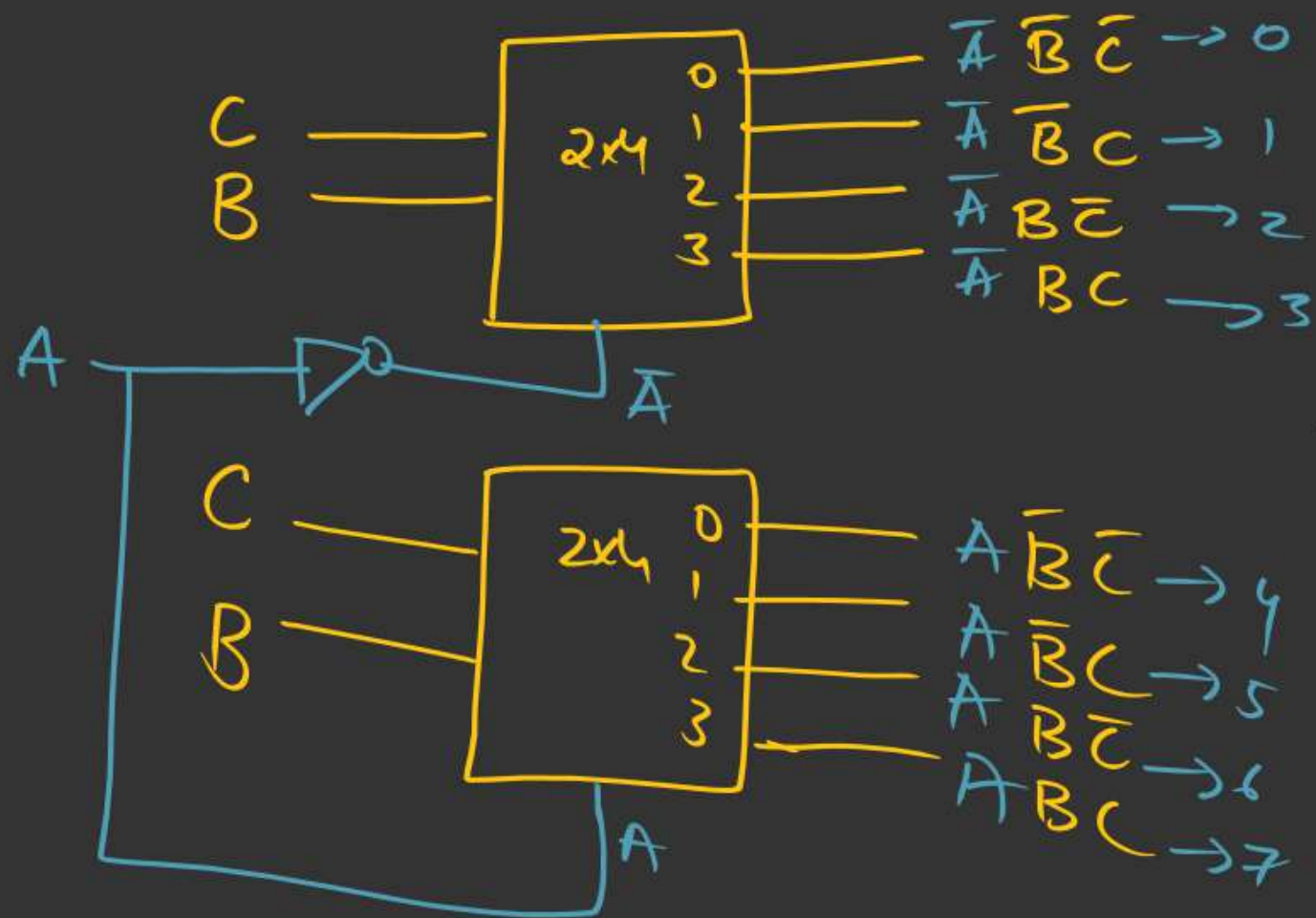
$\} \Rightarrow \Sigma(1, 2, 3)$

(Q.1) How many 2×4 Decoders $\longrightarrow$ 3×8 decoder?

Ans: (2)

$F(A, B, C)$

$2 \times 4 \longrightarrow 3 \times 8$



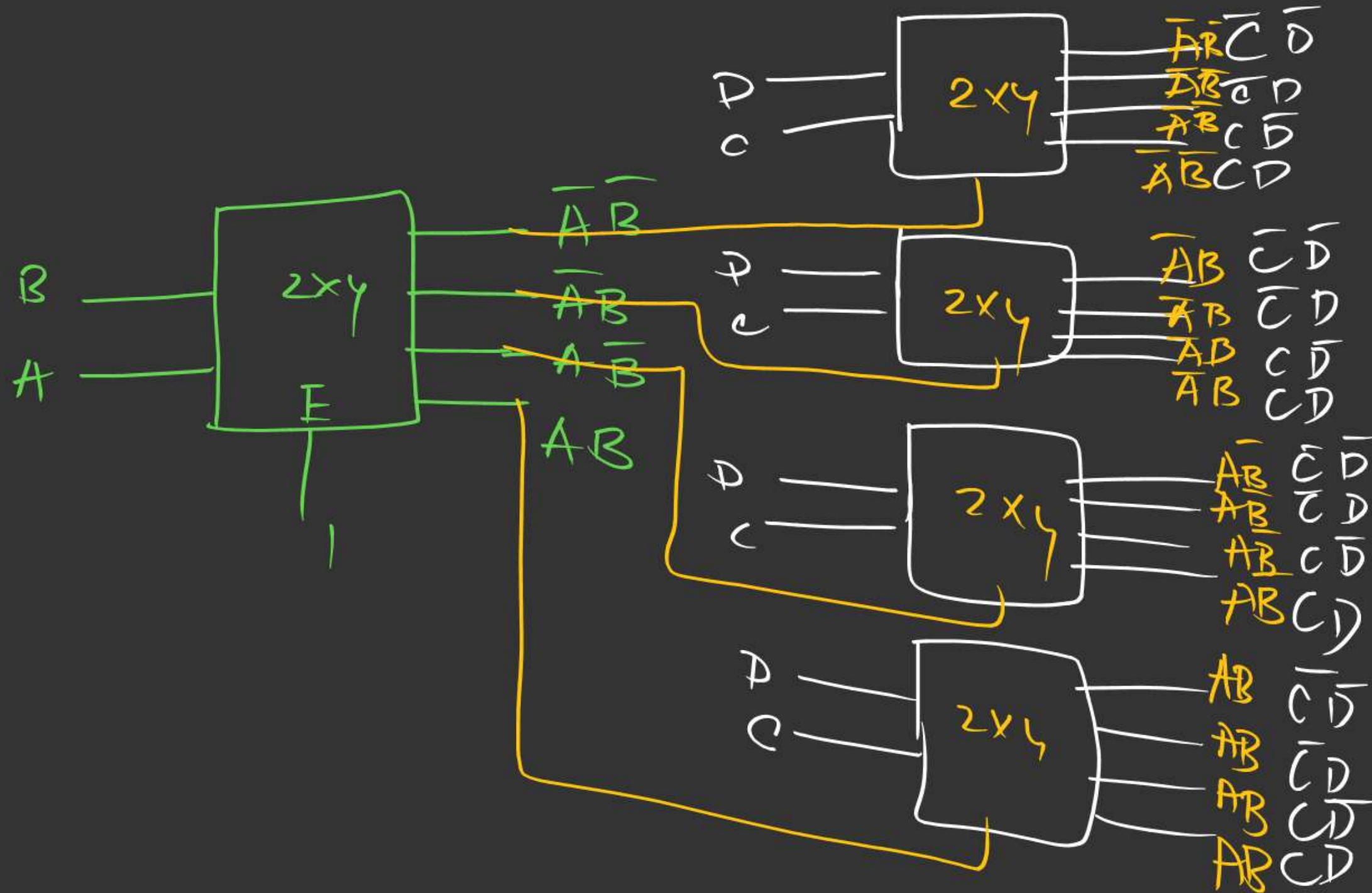
(Q.2) Design

(ABCD)

4x16

using 2x4 decoder

Ans: (5)



Shortcut

1) $\begin{matrix} \textcircled{2} \times 4 \\ \text{dec} \end{matrix} \longrightarrow \begin{matrix} \textcircled{3} \times 8 \\ \text{dec} \end{matrix} \Rightarrow \frac{8}{4} = \textcircled{2} \text{ Ans}$

2) $\begin{matrix} \textcircled{2} \times 4 \\ \text{dec} \end{matrix} \xrightarrow{\times 2} \begin{matrix} \textcircled{4} \times 16 \\ \text{dec} \end{matrix}$

$$\frac{16}{4} = 4$$
$$\frac{4}{4} = 1$$

$$\text{Total} = 4 + 1 = \textcircled{5}$$



(Q3) How many $\textcircled{3} \times 8$ dec $\longrightarrow$ $\textcircled{4} \times 16$ decoder?

$$\frac{16}{8} = \textcircled{2} \checkmark$$

(Q4) $\textcircled{3} \times 8$ dec $\xrightarrow{*2}$ $\textcircled{6} \times 64$? dec

$$\frac{64}{8} = 8$$
$$\frac{8}{8} = 1$$

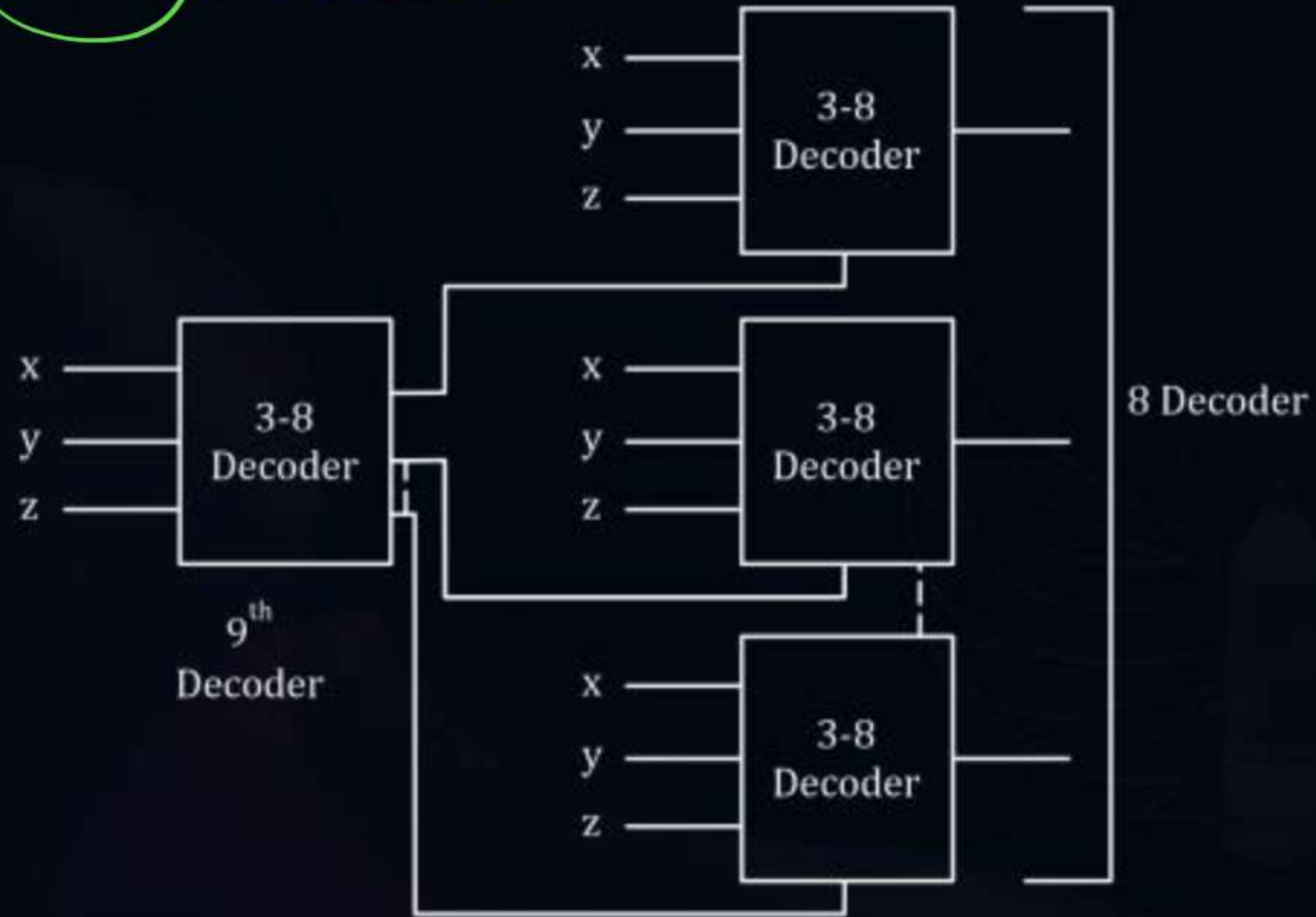
$$\text{Total} = 8 + 1 = 9$$



Topic : DECODER, HA FA

H.W

3 × 8 DECODER → 6 × 64 DECODER



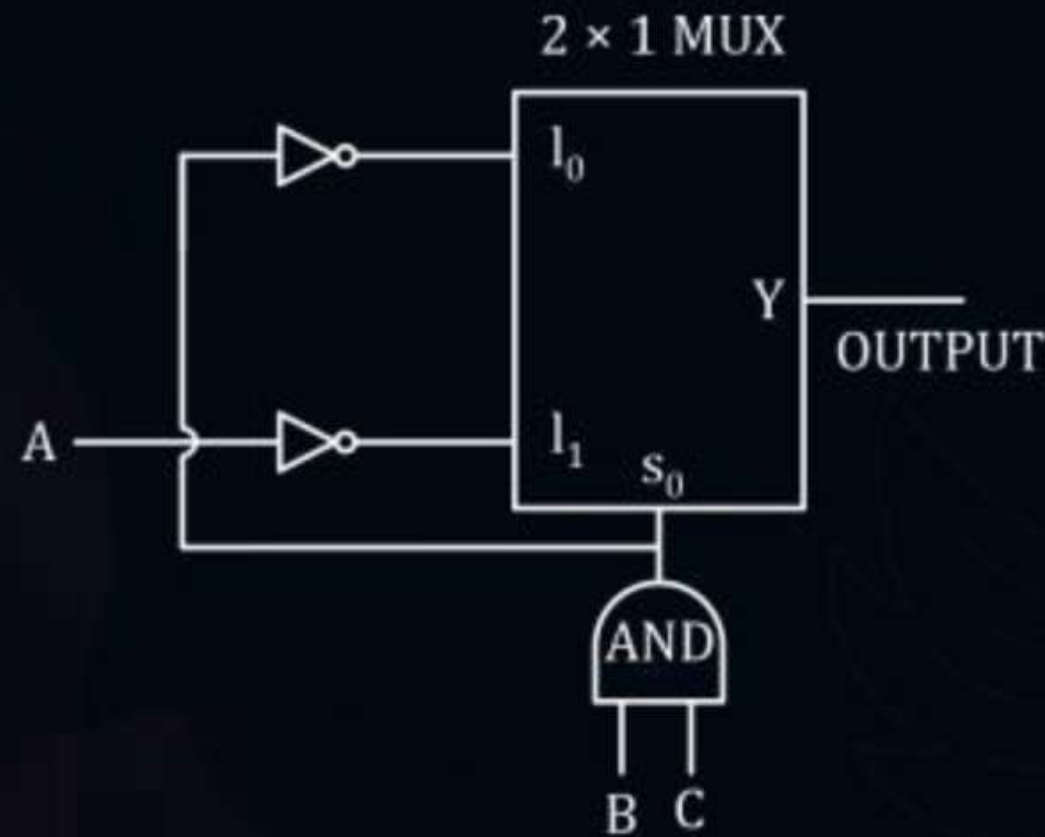
[MCQ]

#Q. The combinational circuit given below implements which of the following:

H.W

60%

- A** NOR gate
- B** XOR gate
- C** NAND gate
- D** None of these





2 mins Summary



→ Decoden
→ Questions



THANK - YOU

